Algebra I Linear Equations Cheatsheet

Module 2

Equation Goal

Use inverse operations to isolate the variable while keeping both sides equal.

$$x + a = b \Rightarrow x = b - a$$

$$ax = b \Rightarrow x = \frac{b}{a} \qquad (a \neq 0)$$

Multi-Step Template

$$a(x + b) + c = dx + e$$

1. Distribute.
2. Combine like terms on each side.
3. Move variable terms to one side.
4. Move constants to the other side.
5. Divide by the coefficient.

Variables on Both Sides

$$3x + 8 = 7x - 12$$

Move variable terms together and constants together:

$$20 = 4x, \qquad x = 5$$

Special Cases

$$5 = 5 \Rightarrow \text{infinitely many solutions}$$

$$5 = 9 \Rightarrow \text{no solution}$$

$$x = 4 \Rightarrow \text{one solution}$$

Literal Equations

Solve for one variable in terms of the others.

$$A = lw \Rightarrow l = \frac{A}{w} \qquad (w \neq 0)$$

$$y = mx + b \Rightarrow x = \frac{y - b}{m} \qquad (m \neq 0)$$

Clearing Fractions

Multiply every term on both sides by the LCD.

$$\frac{x}{3} + \frac{x}{4} = 7$$

$$12\left(\frac{x}{3}\right) + 12\left(\frac{x}{4}\right) = 12(7)$$

Absolute Value Equations

$$|A| = c$$

If

$$c > 0, \quad A = c \text{ or } A = -c$$

If

$$c = 0, \quad A = 0$$

If

$$c < 0, \quad \text{no solution}$$

Quick Checks

- Distribute before moving terms.
- Every operation must be done to both sides.
- Check answers by substitution.
- For absolute value, split into two equations only when the outside value is nonnegative.